Assignment 5

Exploring SQL Grouping, Database Views Implement SQL queries for different SQL concepts such as grouping data, using the GROUP BY clause, employing the HAVING clause, applying the EXISTS/NOT EXISTS operators, Creating and using Database Views.

```
mysql> CREATE DATABASE BankManagement_037;
Query OK, 1 row affected (0.05 sec)

mysql> USE BankManagement_037;
Database changed
mysql> CREATE TABLE Customers (
    ->     customer_id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     dob DATE,
    ->     city VARCHAR(50),
    ->     account_type VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> CREATE TABLE Branches (
    ->     branch_id INT PRIMARY KEY,
    ->     branch_name VARCHAR(50),
    ->     city VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Accounts (
    ->     account_id INT PRIMARY KEY,
    ->     customer_id INT,
    ->     balance DECIMAL(10,2),
    ->     account_type VARCHAR(20),
    ->     branch_id INT,
    ->     FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),
    ->     FOREIGN KEY (branch_id) REFERENCES Branches(branch_id)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Transactions (
    ->     transaction_id INT PRIMARY KEY,
    ->     account_id INT,
    ->     amount DECIMAL(10,2),
    ->     transaction_type VARCHAR(20),
    ->     transaction_date DATE,
    ->     FOREIGN KEY (account_id) REFERENCES Accounts(account_id)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Loans (
    ->     loan_id INT PRIMARY KEY,
    ->     customer_id INT,
    ->     amount DECIMAL(10,2),
    ->     loan_type VARCHAR(20),
    ->     status VARCHAR(20),
    ->     FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO Customers VALUES
    -> (1, 'Amit', '1995-05-10', 'Mumbai', 'Savings'),
    -> (2, 'Neha', '1998-07-15', 'Delhi', 'Current'),
    -> (3, 'Raj', '1992-03-20', 'Mumbai', 'Savings'),
    -> (4, 'Priya', '1996-11-25', 'Chennai', 'Current'),
    -> (5, 'Karan', '1990-09-12', 'Delhi', 'Savings'),
    -> (6, 'Sneha', '1999-01-05', 'Pune', 'Savings'),
    -> (7, 'Arjun', '1993-06-18', 'Pune', 'Current');
Query OK, 7 rows affected (0.03 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Branches VALUES
    -> (1, 'Main Branch', 'Mumbai'),
    -> (2, 'City Branch', 'Delhi'),
    -> (3, 'South Branch', 'Chennai'),
    -> (4, 'West Branch', 'Pune'),
    -> (5, 'Central Branch', 'Mumbai');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Accounts VALUES
    -> (101, 1, 60000, 'Savings', 1),
    -> (102, 2, 30000, 'Current', 2),
    -> (103, 3, 20000, 'Savings', 5),
    -> (104, 4, 80000, 'Current', 3),
    -> (105, 5, 10000, 'Savings', 2),
    -> (106, 1, 40000, 'Savings', 4),
    -> (107, 6, 15000, 'Savings', 4);
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Loans VALUES
    -> (1, 1, 200000, 'Home', 'Approved'),
    -> (2, 2, 150000, 'Car', 'Pending'),
    -> (3, 3, 50000, 'Personal', 'Approved'),
    -> (4, 4, 300000, 'Home', 'Rejected'),
    -> (5, 6, 100000, 'Car', 'Approved'),
    -> (6, 7, 120000, 'Home', 'Approved');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Transactions VALUES
    -> (7, 102, 30000, 'Deposit', '2025-03-01'),
    -> (8, 103, 4000, 'Deposit', '2025-03-01'),
    -> (9, 104, 6000, 'Withdrawal', '2025-03-01'),
    -> (10, 105, 8000, 'Transfer', '2025-03-01'),
    -> (11, 101, 25000, 'Deposit', '2025-03-05'),
    -> (12, 102, 35000, 'Deposit', '2025-03-05');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
mysql> SHOW TABLES;
+-----------------------------+
| Tables_in_bankmanagement_037 |
+-----------------------------+
| accounts                    |
| branches                    |
| customers                   |
| loans                       |
| transactions                |
+-----------------------------+
5 rows in set (0.04 sec)
```

```
mysql> SELECT * FROM Customers;
+-------------+-------+------------+---------+--------------+
| customer_id | name  | dob        | city    | account_type |
+-------------+-------+------------+---------+--------------+
|           1 | Amit  | 1995-05-10 | Mumbai  | Savings      |
|           2 | Neha  | 1998-07-15 | Delhi   | Current      |
|           3 | Raj   | 1992-03-20 | Mumbai  | Savings      |
|           4 | Priya | 1996-11-25 | Chennai | Current      |
|           5 | Karan | 1990-09-12 | Delhi   | Savings      |
|           6 | Sneha | 1999-01-05 | Pune    | Savings      |
|           7 | Arjun | 1993-06-18 | Pune    | Current      |
+-------------+-------+------------+---------+--------------+
7 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Accounts;
+------------+-------------+----------+--------------+-----------+
| account_id | customer_id | balance  | account_type | branch_id |
+------------+-------------+----------+--------------+-----------+
|        101 |           1 | 60000.00 | Savings      |         1 |
|        102 |           2 | 30000.00 | Current      |         2 |
|        103 |           3 | 20000.00 | Savings      |         5 |
|        104 |           4 | 80000.00 | Current      |         3 |
|        105 |           5 | 10000.00 | Savings      |         2 |
|        106 |           1 | 40000.00 | Savings      |         4 |
|        107 |           6 | 15000.00 | Savings      |         4 |
+------------+-------------+----------+--------------+-----------+
7 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Transactions;
+----------------+------------+----------+------------------+------------------+
| transaction_id | account_id | amount   | transaction_type | transaction_date |
+----------------+------------+----------+------------------+------------------+
|              7 |        102 | 30000.00 | Deposit          | 2025-03-01       |
|              8 |        103 |  4000.00 | Deposit          | 2025-03-01       |
|              9 |        104 |  6000.00 | Withdrawal       | 2025-03-01       |
|             10 |        105 |  8000.00 | Transfer         | 2025-03-01       |
|             11 |        101 | 25000.00 | Deposit          | 2025-03-05       |
|             12 |        102 | 35000.00 | Deposit          | 2025-03-05       |
+----------------+------------+----------+------------------+------------------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Branches;
+------------+-----------------+---------+
| branch_id | branch_name     | city    |
+------------+-----------------+---------+
|          1 | Main Branch     | Mumbai  |
|          2 | City Branch     | Delhi   |
|          3 | South Branch    | Chennai |
|          4 | West Branch     | Pune    |
|          5 | Central Branch  | Mumbai  |
+------------+-----------------+---------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Loans;
+---------+-------------+-----------+-----------+----------+
| loan_id | customer_id | amount    | loan_type | status   |
+---------+-------------+-----------+-----------+----------+
|       1 |           1 | 200000.00 | Home      | Approved |
|       2 |           2 | 150000.00 | Car       | Pending  |
|       3 |           3 |  50000.00 | Personal  | Approved |
|       4 |           4 | 300000.00 | Home      | Rejected |
|       5 |           6 | 100000.00 | Car       | Approved |
|       6 |           7 | 120000.00 | Home      | Approved |
+---------+-------------+-----------+-----------+----------+
6 rows in set (0.00 sec)
```

1. Retrieve the total balance for each account type.

```
mysql> SELECT account_type, SUM(balance) AS total_balance
    -> FROM Accounts
    -> GROUP BY account_type;
+--------------+---------------+
| account_type | total_balance |
+--------------+---------------+
| Savings      |     145000.00 |
| Current      |     110000.00 |
+--------------+---------------+
2 rows in set (0.01 sec)
```

2. Count the number of accounts in each branch.

```
mysql> SELECT branch_id, COUNT(*) AS total_accounts
    -> FROM Accounts
    -> GROUP BY branch_id;
+-----------+----------------+
| branch_id | total_accounts |
+-----------+----------------+
|         1 |              1 |
|         2 |              2 |
|         3 |              1 |
|         4 |              2 |
|         5 |              1 |
+-----------+----------------+
5 rows in set (0.00 sec)
```

3. Find the number of customers in each city.

```
mysql> SELECT city, COUNT(*) AS total_customers
    -> FROM Customers
    -> GROUP BY city;
+---------+-----------------+
| city    | total_customers |
+---------+-----------------+
| Mumbai  |               2 |
| Delhi   |               2 |
| Chennai |               1 |
| Pune    |               2 |
+---------+-----------------+
4 rows in set (0.00 sec)
```

4. Show the number of loans approved per loan type.

```
mysql> SELECT loan_type, COUNT(*) AS approved_loans
    -> FROM Loans
    -> WHERE status = 'Approved'
    -> GROUP BY loan_type;
+-----------+----------------+
| loan_type | approved_loans |
+-----------+----------------+
| Home      |              2 |
| Personal  |              1 |
| Car       |              1 |
+-----------+----------------+
3 rows in set (0.00 sec)
```

5. Find the total number of transactions for each transaction type.

```
mysql> SELECT transaction_type, COUNT(*) AS total_transactions
    -> FROM Transactions
    -> GROUP BY transaction_type;
+------------------+--------------------+
| transaction_type | total_transactions |
+------------------+--------------------+
| Deposit          |                  4 |
| Withdrawal       |                  1 |
| Transfer         |                  1 |
+------------------+--------------------+
3 rows in set (0.00 sec)
```

6. Show the customers who do not have a loan

```
mysql> SELECT * FROM Customers c
    -> WHERE NOT EXISTS (
    ->     SELECT 1 FROM Loans l WHERE l.customer_id = c.customer_id
    -> );
+-------------+-------+------------+-------+--------------+
| customer_id | name  | dob        | city  | account_type |
+-------------+-------+------------+-------+--------------+
|           5 | Karan | 1990-09-12 | Delhi | Savings      |
+-------------+-------+------------+-------+--------------+
1 row in set (0.00 sec)
```

7. Retrieve customers who have accounts in more than one branch.

```
mysql> SELECT customer_id
    -> FROM Accounts
    -> GROUP BY customer_id
    -> HAVING COUNT(DISTINCT branch_id) > 1;
+--------------+
| customer_id |
+--------------+
|            1 |
+--------------+
1 row in set (0.00 sec)
```

8. Find all accounts that have not had any deposits in the last 3 months.

```
mysql> SELECT * FROM Accounts a
    -> WHERE NOT EXISTS (
    ->     SELECT 1 FROM Transactions t
    ->     WHERE t.account_id = a.account_id
    ->     AND t.transaction_type = 'Deposit'
    ->     AND t.transaction_date >= DATE_SUB(CURDATE(), INTERVAL 3 MONTH)
    -> );
+------------+-------------+----------+--------------+-----------+
| account_id | customer_id | balance  | account_type | branch_id |
+------------+-------------+----------+--------------+-----------+
|        101 |           1 | 60000.00 | Savings      |         1 |
|        102 |           2 | 30000.00 | Current      |         2 |
|        103 |           3 | 20000.00 | Savings      |         5 |
|        104 |           4 | 80000.00 | Current      |         3 |
|        105 |           5 | 10000.00 | Savings      |         2 |
|        106 |           1 | 40000.00 | Savings      |         4 |
|        107 |           6 | 15000.00 | Savings      |         4 |
+------------+-------------+----------+--------------+-----------+
7 rows in set (0.00 sec)
```

9. Display the account types where the total balance is below ₹25,000.

```
mysql> SELECT account_type, SUM(balance) AS total_balance
    -> FROM Accounts
    -> GROUP BY account_type
    -> HAVING SUM(balance) < 25000;
Empty set (0.00 sec)
```

10. Find the total balance per account type where the balance is above ₹50,000.

```
mysql> SELECT account_type, SUM(balance) AS total_balance
    -> FROM Accounts
    -> GROUP BY account_type
    -> HAVING SUM(balance) > 50000;
+--------------+---------------+
| account_type | total_balance |
+--------------+---------------+
| Savings      |     145000.00 |
| Current      |     110000.00 |
+--------------+---------------+
2 rows in set (0.00 sec)
```

11. Retrieve the count of transactions made on dates where more than 5 transactions occurred.

```
mysql> SELECT transaction_date, COUNT(*) AS total_transactions
    -> FROM Transactions
    -> GROUP BY transaction_date
    -> HAVING COUNT(*) > 5;
Empty set (0.00 sec)
```

12. Find the top 3 transaction days with the highest total transaction amount.

```
mysql> SELECT transaction_date, SUM(amount) AS total_amount
    -> FROM Transactions
    -> GROUP BY transaction_date
    -> ORDER BY total_amount DESC
    -> LIMIT 3;
+------------------+--------------+
| transaction_date | total_amount |
+------------------+--------------+
| 2025-03-05       |     60000.00 |
| 2025-03-01       |     48000.00 |
+------------------+--------------+
2 rows in set (0.00 sec)
```

13. Find customers who have a loan but no account in the bank.

```
mysql> SELECT * FROM Customers c
    -> WHERE EXISTS (
    ->     SELECT 1 FROM Loans l WHERE l.customer_id = c.customer_id
    -> )
    -> AND NOT EXISTS (
    ->     SELECT 1 FROM Accounts a WHERE a.customer_id = c.customer_id
    -> );
+-------------+-------+------------+------+--------------+
| customer_id | name  | dob        | city | account_type |
+-------------+-------+------------+------+--------------+
|           7 | Arjun | 1993-06-18 | Pune | Current      |
+-------------+-------+------------+------+--------------+
1 row in set (0.00 sec)
```

14. Create a view named ***Customer_Account_Details*** that displays customer name, account ID, account type, balance, and branch name.Using the view ***Customer_Account_Details*** retrieve Data from the View like Display customers whose account balance is greater than ₹50,000.

```
mysql> CREATE VIEW Customer_Account_Details AS
    -> SELECT c.customer_id, c.name, a.account_id, a.account_type, a.balance, b.branch_name
    -> FROM Customers c
    -> JOIN Accounts a ON c.customer_id = a.customer_id
    -> JOIN Branches b ON a.branch_id = b.branch_id;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT name, account_id, account_type, balance, branch_name
    -> FROM Customer_Account_Details
    -> WHERE balance > 50000;
+-------+------------+--------------+----------+--------------+
| name  | account_id | account_type | balance  | branch_name  |
+-------+------------+--------------+----------+--------------+
| Amit  |        101 | Savings      | 60000.00 | Main Branch  |
| Priya |        104 | Current      | 80000.00 | South Branch |
+-------+------------+--------------+----------+--------------+
2 rows in set (0.00 sec)
```

15.**Create a view** named *Transaction_Summary* that shows account ID, transaction type, total number of transactions, and total transaction amount.**Write a query to retrieve** accounts from the view *Transaction_Summary* where the total transaction amount is greater than ₹1,00,000.

```
mysql> CREATE VIEW Transaction_Summary AS
    -> SELECT account_id, transaction_type,
    -> COUNT(*) AS total_transactions,
    -> SUM(amount) AS total_amount
    -> FROM Transactions
    -> GROUP BY account_id, transaction_type;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Transaction_Summary
    -> WHERE total_amount > 100000;
Empty set (0.00 sec)
```

16.Create a view that displays customer name and loan details for all customers whose loan status is Approved.Write a query to display the customers from the view who have taken a Home loan.

```
mysql> CREATE VIEW Approved_Loan_Customers AS
    -> SELECT c.customer_id, c.name, l.loan_id, l.loan_type, l.amount, l.status
    -> FROM Customers c
    -> JOIN Loans l ON c.customer_id = l.customer_id
    -> WHERE l.status = 'Approved';
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Approved_Loan_Customers
    -> WHERE loan_type = 'Home';
+-------------+-------+---------+-----------+-----------+----------+
| customer_id | name  | loan_id | loan_type | amount    | status   |
+-------------+-------+---------+-----------+-----------+----------+
|           1 | Amit  |       1 | Home      | 200000.00 | Approved |
|           7 | Arjun |       6 | Home      | 120000.00 | Approved |
+-------------+-------+---------+-----------+-----------+----------+
2 rows in set (0.00 sec)
```